Enrollment No.: _____

Darshan UNIVERSITY

योगः कर्मसु कौशलम्

Darshan Institute of Computer Application

B.Sc. (IT) | Sem-3 | Summer-2023

Course Code	:	2106CS301	**Date**	:	24-04-2023
Course Name	:	Computer Network and Network Security	**Duration**	:	150 Minutes
			Total Marks	:	70

Instructions:

1. Attempt all the questions.
2. Figures to the right indicates maximum marks.
3. Make suitable assumptions wherever necessary.

Q.1	**(A)**	Define TCP and UDP.	**4**
	(B)	List the functions of Data-link layer.	**3**
		OR	
		List the functions of Network layer.	
	(C)	Write a short note on TCP/IP Model with diagram.	**7**
		OR	
		Describe the Functions of the Transport Layer.	
Q.2	**(A)**	Write Advantages and Disadvantages of FDM.	**4**
	(B)	Define Circuit Switching.	**3**
		OR	
		Write a full form of FDM, TDM, WDM.	
	(C)	List the multiplexing techniques, and define anyone in detail.	**7**
		OR	
		List the types of Switching and explain anyone in brief.	
Q.3	**(A)**	Calculate how many hosts can you have for create the network. (You have been allocated a class C network address of 211.1.1.0 and are using the default subnet mask of 255.255.255.0)	**4**
	(B)	Describe Uniform Resource Locator	**3**
		OR	
		Classify IP Addresses with example.	
	(C)	Demonstrate subnetting in detail.	**7**
		OR	

Given IP Address – 172.16.0.0/25, Calculate the number of subnets and the number of hosts per subnet. Also, for the first subnet block, calculate the subnet address, first host ID, last host ID and broadcast address.

Q.4	**(A)**	List the parameters used for calculating a routing metric.	**4**
	(B)	Explain the direct broadcast and limited broadcast.	**3**
		OR	
		Describe Unicast and Broadcast.	
	(C)	Discuss the Adaptive Routing algorithm and Non-adaptive Routing algorithm.	**7**
		OR	
		Explain Shortest Path Routing.	
Q.5	**(A)**	Clarify Authentication and Authorization.	**4**
	(B)	List the useful Security Policies.	**3**
		OR	
		List the Security Methods.	
	(C)	Explain Encryption and Decryption in detail.	**7**
		OR	
		Explain Virtual Private Network (VPN) in detail.	

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